RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY

COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT - 3

Prepared by: Mohamed. B. Sirajuddeen

Reg No : 22TD0053

Subject Name : Object Oriented Analysis And Design

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Unit : 3

Teacher's Sign :

Department Vision And Mission

Vision

Cultivate student as technocrats, researchers and entrepreneurs in the field of computer science and engineering.

Mission

→ The impart quality education in computer science and Engineering by means of learning technique and value-added courses.

→ To inculcate professionals excellence and research culture by encouraging projects in cutting edge technologies through industry instruction.

→ To build leadership skill ethical values and team work among the students.

→ To strengthen the collaboration of department and industry internship mentorships and professional body activities.

2 marks

1. What are the processes of object-oriented design?
   - → Requirements gathering
   - → Analysis
   - → Design
   - → Object-oriented design principles
   - → Class and object design
   - → System architecture design
   - → Implementation.
   - → Testing and validation
   - → Maintenance and evolution.

2. Outline the two major properties of a-part-of relationship?

Two major properties of a-part-of relationship are:
- → transitivity
- → Antisymmetry.

Transitivity If A is part of B and B is part of C, then A is part of C.

Eg: A carburetor is part of an engine and an engine is part of a car; therefore a carburetor is part of a car.

Antisymmetry If A is part of B, then B is not part of A.

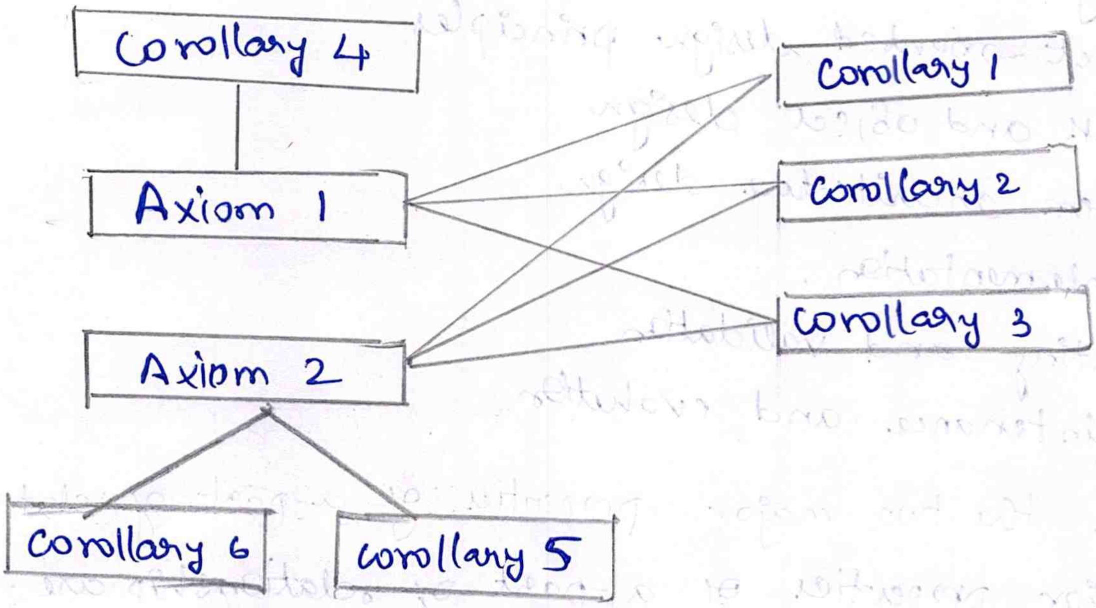
Eg: An engine is part of a car; but a car is not a part of an engine.

3. State the significance of design axioms and corollary.

Axioms An axiom is a fundamental truth that always is observed to be valid and for which there is no counter example or exception.

Axiom 1 : The independence axiom. Maintain the independence of components.

Axiom 2 : The information axiom : Minimize the information content of the design.



4. State the purpose of user defined compartments in class diagram

→ Enhance clarity

→ Add custom details

→ Improve communication

→ Support design decisions

5. Differentiate generalization and specialization.

S.No	Generalization	Specialization.
1.	The process of extracting shared characteristics from zero or more classes and combining them into a generalized super class.	The process of creating new subclasses from an existing class by adding unique attributes or behavior
2.	To promote code reuse and abstraction by identifying common features	To provide specific implementation for a more general class.
3.	Bottom-up (from subclasses to superclass)	Top-down (from superclass to subclasses)

6. Define Modular design.

Modular design is the process of breaking down a complex system into smaller, manageable and reusable modules (often represented as classes or components) each with well defined interfaces and responsibilities.

→ Encapsulation.
→ Cohesion
→ low - coupling
→ Reusability
→ Maintainability.

7. State the importance of design pattern.

→ Reusability
→ Standardized solutions.
→ Improved design quality.
→ Ease of maintenance and scalability.
→ Faster development.
→ Better documentation and understanding.
→ Facilitates object-oriented principles.

8. Explain what is common class pattern strategy.

The common class pattern strategy is a method to find potential classes by examining common pattern of responsibilities and roles that objects typically play in a system.

class pattern

→ Entity classes or model classes
→ Boundary classes or interface classes
→ control classes.

9. What is irrelevant classes? Give example.

→ Irrelevant classes are classes that donot contribute meaningfully to the system's objectivies or functionality in object-oriented analysis and design.

→ Including them can lead to unnecessary complexity cluttered design and poor maintainability.

Example

Book Author Birthday.
although author might be relevant, tracking the author's birthday may not serve any purpose in the systems. unless there is a specific use case like send birthday wishes to authors, which is highly unlikely in a library system. So, BookAuthorBirthday would be irrelevant unless justified by a requirement.

10. What is a usecase? List the questions to identify a usecase.

→ A usecase in object - oriented analysis and design is a description of a system's behaviours as it responds to a request from an external actor (like a user or another system).

→ It represents a functional requirement and describes how the system interacts with uses to achieve a specific goal.

Question to Identify a Use case:

1. Who will use the system?
2. What goals does each actor want to accomplish?
3. What tasks or activities will the system perform in response?
4. What are the main functions the system must provide?
5. What information will the actor input or receive?

Part - B

11. With a suitable example explain how to design a class. give all possible representation in a class (name, attributes, visibility, methods and responsibilities.

→ In an object-oriented environment, objects take on an active role in a system. Of course, objects do not exist in isolation but interact with each other.

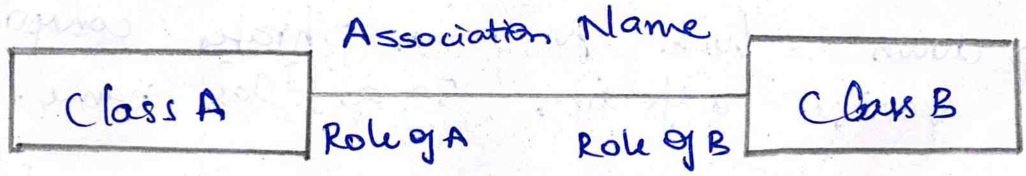
→ The relationship among objects is based on the assumption each makes about the other object, including what operations can be performed and behaviour results.

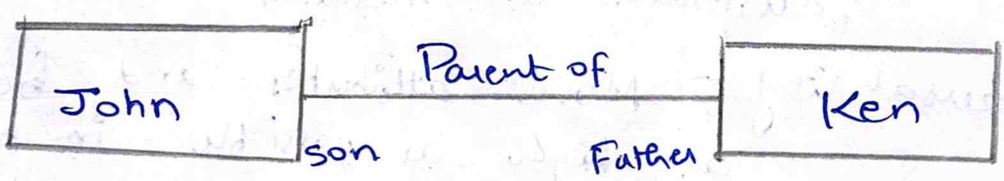
There types of relationships among objects are,

- Association
- Super sub structure
- Aggregation and a-part-of structure.

Associations :-

Association represents a physical or conceptual connection between two or more objects.





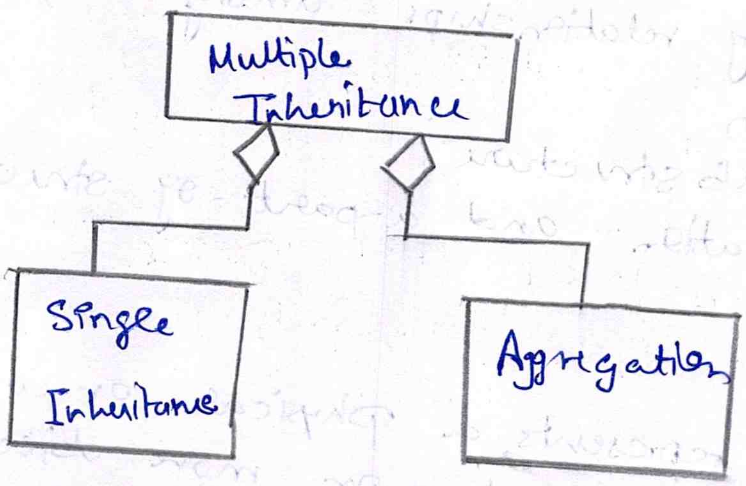
Guideline for Identifying Association:

→ A dependency between two or more classes may be an association

→ A reference from one class to another is an association. some association are implicit or taken from general knowledge.

Super-Sub class relationships

Super class - sub class relationships - also known as generalization hierarchy, allow objects to be built from other objects. Such relationships are exiplicit advantage to the commonality of objects.



Guidelines for Identifying Super - sub relationship.

→ Top down - look for noun phrases composed of various adjectives in a class name.

→ Bottom up - look for classes with similar attributes or methods.

→ Reusability - More attributes and behaviors as high as possible in the hierarchy.

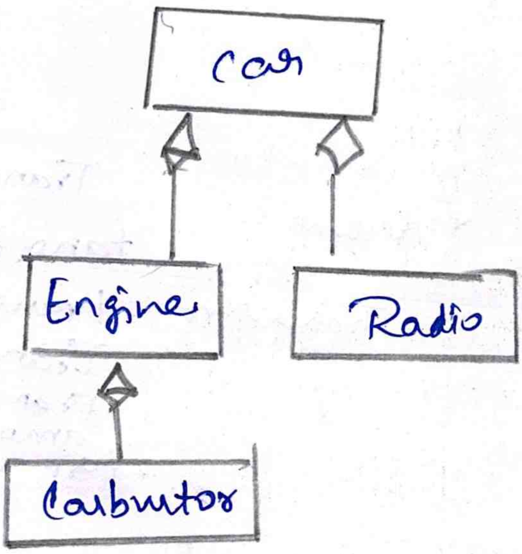
A-part-of relationship:

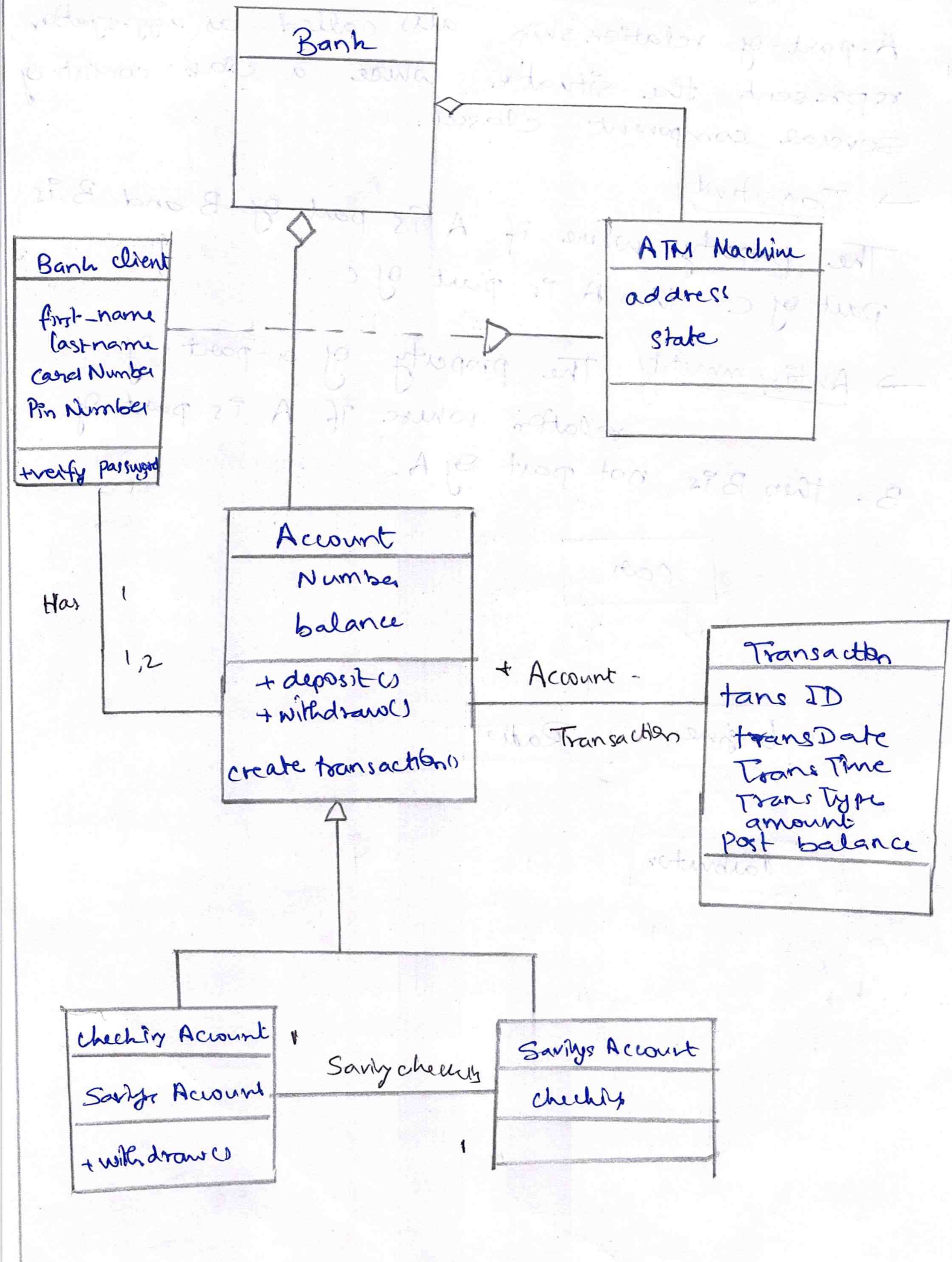
A-part-of relationship also called as aggregation, represent the situation where a class consists of several component classes.

→ Transitivity

The property where if A is part of B and B is part of C, then A is part of C.

→ Antisymmetry The property of a-part-of relation where if A is part of B. then B is not part of A.




Bank
Bank client
first_name
lastname
Card Number
Pin Number
+verify password
ATM Machine
address
State
Has
1
1,2
Account
Number
balance
+ deposit()
+ withdraw()
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12. List all approaches introduced for identifying classes in a domain and explain any three of them.

- The four alternative approaches for identifying classes, the noun phrase approach, the common class patterns approach, the use case driven, sequence collaboration modelling approach and the classes, responsibilities and collaboration (CRC) approach.

- The mov technique than method is uses for identifying classes responsibilies and therefore their attributes and methods.

1. <u>Noun</u> - phr<u>ase</u> appr<u>oach</u>

- The noun-phrase approach was proposed by Rebecca Wirfs-brock. Brian wilker and his co-workers.

In this method, read through the requirements or use cases looking for noun phrases.

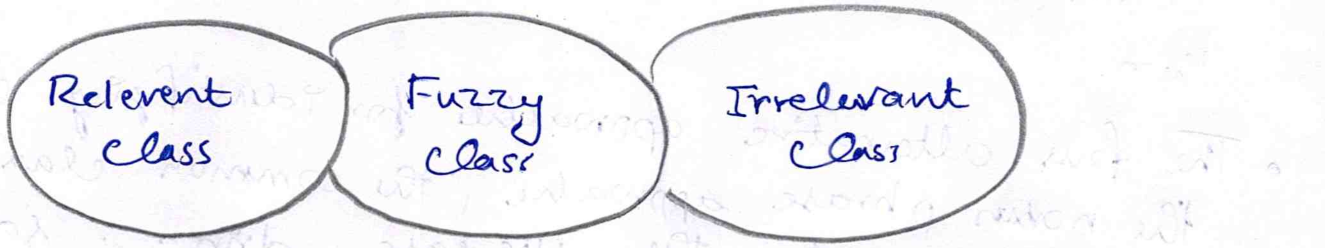
- All plurals are changed to singular, the nouns are listed and the list divided into three categories.
- Depending on whether such object modeling is for the analysis or design phase of development, some classes may need to be added or removed from the model and remember flexibility is a virtue.

<u>Identifying</u> Tentative <u>classes</u>

→ Look for nouns and noun phrases in the use case

→ Some classes are implicit or taken from general knowledge.

→ Carefully choose and define class names.



Selecting classes from the relevant and Fuzzy categories

Redundant classes: Do not keep two classes that express the same information. If more than one word is being used to describe the same idea, select the one that is the most meaningful in the context of the system.

Adjective class Adjective can be used in many ways. An adjective can suggest a different kind of object, different use of the same object or it could be utterly irrelevant.

Example Adult members behave differently than youth members. So the two should be classified as different classes.

Attribute classes Tentative objects that are used only as value should be defined or restarted as attributes and not as a class.

Example Client status and demographic of client are not classes but attributes of the client class.

Irrelevant classes

Each class must have a purpose and every class should be clearly defined and necessary. You must formulate a statement of purpose for each candidate class.

2. Common - class patterns Approach

The second method for identifying classes is using common class pattern, which is based on a knowledge base of the common class that have been proposed by various researches. They have compiled and listed in following patterns for finding the candidate class and object.

Name : Concept class

Context : A concept is a particular idea or understanding that we have of our world.

The concept class encompasses principles that are not tangible but used to organize or keeps track of business activities or communications. privately held ideas or notions are called conceptions. when an understanding is shared by another, it becomes a concept. To communicate with others, we must share our individually held conceptions and arrives at agreed concepts.

Eg: Performance is an example of concept class object.

Name: Event class

Context: Events classes are points in time that must be recorded. Things happen, usually to something else at a given date and time or as a step in an ordered sequence.

Eg: Landing, interrupt, request and order to possible events.

Name : Organization class

Context: An organization class is a collection of people ~~resources~~, facilities, or groups to which the user belongs their capabilities have a defined mission.

Eg: An accounting department might be considered a potential class.

Name: people class

The people class represent the different roles users play in interacting with the application. people carry out some function.

Eg: Employee, client, teacher, and manage are examples of people.

Name: places class

Places are physical locations that the system must keep information about.

Eg: Building, store, sites and offices are examples of places.

Name: Tangible things and devices class.

This class includes physical objects or group of objects that are tangible and devices.

Eg: Cars are an example of tangible things and pressure sensor are an example of devices

3. Classes, Responsibilities And collaboration

- The CRC can help us identify classes CRC is more a teaching technique than a method for identifying classes.

- CRC is based on the idea that can object either can accomplish a certain responsibility itself or it may require the assitance of other objects.
- It it requires the assitance of other objects, it must collaborate with those objects to fulfil its responsibility.
- CRC cards are $4" \times 6"$ index cards all the information for an object is writen on a card, which is cheap portable, readilly available and familiar.
- The class name should appear in the upper left-hand corner. a list of responsibilities should appear under it in the left two thirds of the card in the left two thirds of the card.
- However, rather than simply tracking the details of a collaboration in the form of message sending designers focus on the motivation for respresenting many messages or phrases in english text.

Class Name	Collaborators
Responsibilities	* * *
* * *	

CRC process

CRC process consists of three steps

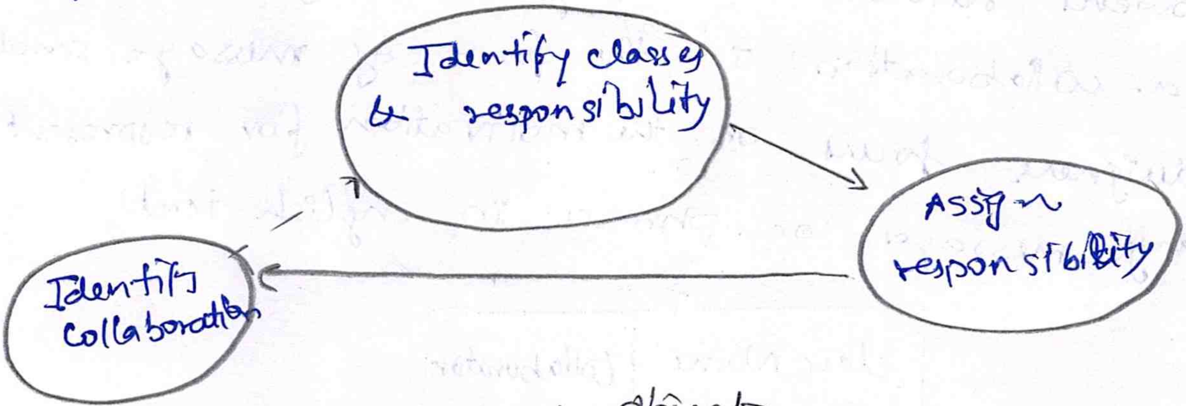
1. Identify classes responsibilities
2. Assign responsibilities.

3. Identify collaboration.

- Classes are identified and grouped by common attributes, which also provides candidates for super class. The class names then are written onto classes, responsibilities and collaborations cards.

- The card also notes sub and superclass to show the class structure. The application requirements then are examined for actions and information associated with each ~~ato~~ class to find the responsibilities of each class.

- Next the responsibilities are distributed. They should be as general as possible and placed as high as possible in the inheritance hierarchy.



CRC for the account object

Account balance number - - - deposit withdraw getBalance	Checking Account (sub class) Savings Account (sub class) Transaction.